

Week 8-10 - DSA Practice Tasks

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For Beginners – Hands-on Practice

Topics: Searching, Stack, Queue, Linked List

Language: C

Searching

1. Linear Search in Array

Problem: Given an array of integers and a target value, find the index of the target element. If not found, return `-1`.

Practice Link: <https://practice.geeksforgeeks.org/problems/search-an-element-in-an-array-1587115621/1>

2. Count Occurrences of an Element

Problem: Count how many times a given element appears in an array.

Practice Link:

<https://practice.geeksforgeeks.org/problems/count-the-occurrences-of-an-element-in-a-sorted-array/1>

3. Binary Search

Problem: Given a sorted array, implement binary search to find a target element.

Practice Link: <https://leetcode.com/problems/binary-search/>

Stack

4. Implement Stack Using Array

Problem: Implement stack operations: push, pop, and display using an array.

Practice Link: <https://practice.geeksforgeeks.org/problems/implement-stack-using-array/1>

5. Valid Parentheses

Problem: Check whether a given string of parentheses is balanced using a stack.

Practice Link: <https://leetcode.com/problems/valid-parentheses/>

6. Reverse String Using Stack

Problem: Reverse a given string using stack operations.

Practice Link: <https://practice.geeksforgeeks.org/problems/reverse-a-string-using-stack/1>



Queue

7. Implement Queue Using Array

Problem: Implement enqueue, dequeue, and display operations using an array.

Practice Link: <https://practice.geeksforgeeks.org/problems/implement-queue-using-array/1>

8. Reverse a Queue

Problem: Reverse a queue using a stack.

Practice Link: <https://practice.geeksforgeeks.org/problems/reverse-a-queue/1>



Linked List

9. Search in Singly Linked List

Problem: Search for a given value in a singly linked list.

Practice Link: <https://practice.geeksforgeeks.org/problems/search-in-linked-list-1664434326/1>

10. Reverse a Singly Linked List

Problem: Reverse a singly linked list and print the reversed list.

Practice Link: <https://leetcode.com/problems/reverse-linked-list/>